

Cloud Workload Characterization Transcript

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Howdy, My name is Robert Lightfoot. Welcome to this module in the CSCE 412 series, Cloud Computing.

Cloud Workload Characterization

In this video, we will be discussing some of the different types of cloud workloads, the bottlenecks associated with the workloads, and how providers address and optimize for the various workloads.



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Cloud Workload Characterization

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What workloads can utilize Cloud Computing?

- Categories
 - Streaming Data
 - Database Creations, Calculations, Search, and Access
 - Data Storage
 - In-Memory Database
 - Ants (Many tiny tasks)
 - High-Performance Compute Calculations
 - Tightly Coupled
 - Separable
 - Location-Aware and Geographically Dispersed
 - Access Control
 - VoIP

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While the cloud can be used for general purpose work, it is typically more well suited and can be better customized for specific workloads. This can be for things like streaming big data, which prioritizes a larger network bandwidth over aspects such as raw CPU performance, available system memory, or even network latency.

On the other hand, workloads such as database calculations would prioritize CPU performance and, if large enough, sizable amounts of memory over general network performance. Another, more recent workload is machine learning or artificial intelligence training, which is offered by providers such as Google's Cloud under their AI Platform.

This workload leverages GPUs, or even special built processors, and their high number of arithmetic logic units to perform parallelized computations, where CPUs typically cannot as they are more equipped to perform complex logic in a sequential fashion. Since there are several other workloads that utilize different combinations of resources, we see most cloud providers offering customizability of the cloud instances available to order. This allows customers to pick and choose which resources they need and which they can sacrifice.

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What metrics should be monitored to determine and alleviate bottlenecks?

- IPC (Instructions per Cycle)
- LLC (Last Level Cache) Misses
- L1 (Level 1) Cache Misses
- Instruction Queue is Full
- Reservation Stations Full

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In some instances, it may be hard to determine the kind of resource allocations to make for a given service or application, leaving the potential for excess overhead or a bottleneck of resources. This makes monitoring system resources crucial for optimizing the system to better meet customer needs. When monitoring performance, aspects such as Instructions per Cycle (IPC), Level 1 (L1) cache misses, and amount fetched from memory are good indicators of performance impacts.

In cases where the IPC is not being fully utilized, the CPU can be shown as running at 100% yet leave performance on the table. An analogy of this would be if a truck driver were to load a truck at half of its capacity each time but still deliver to the same place every time, thus wasting time. If a system were to experience L1 cache misses, this would require the CPU to retrieve data from slower locations such as L2 or L3 cache, which wastes more time and space.

Additionally, if a program needs to access a lot of data, accessing the memory is far slower than any level of cache, thus causing even further slowdowns. However, should there be more data than the memory can hold, the system would then move to an even slower location of data, such as a solid-state drive or hard drive. Upon recognizing problems within the resource utilization, instances can be reconfigured with different allocations or migrated to a more suitable system altogether.

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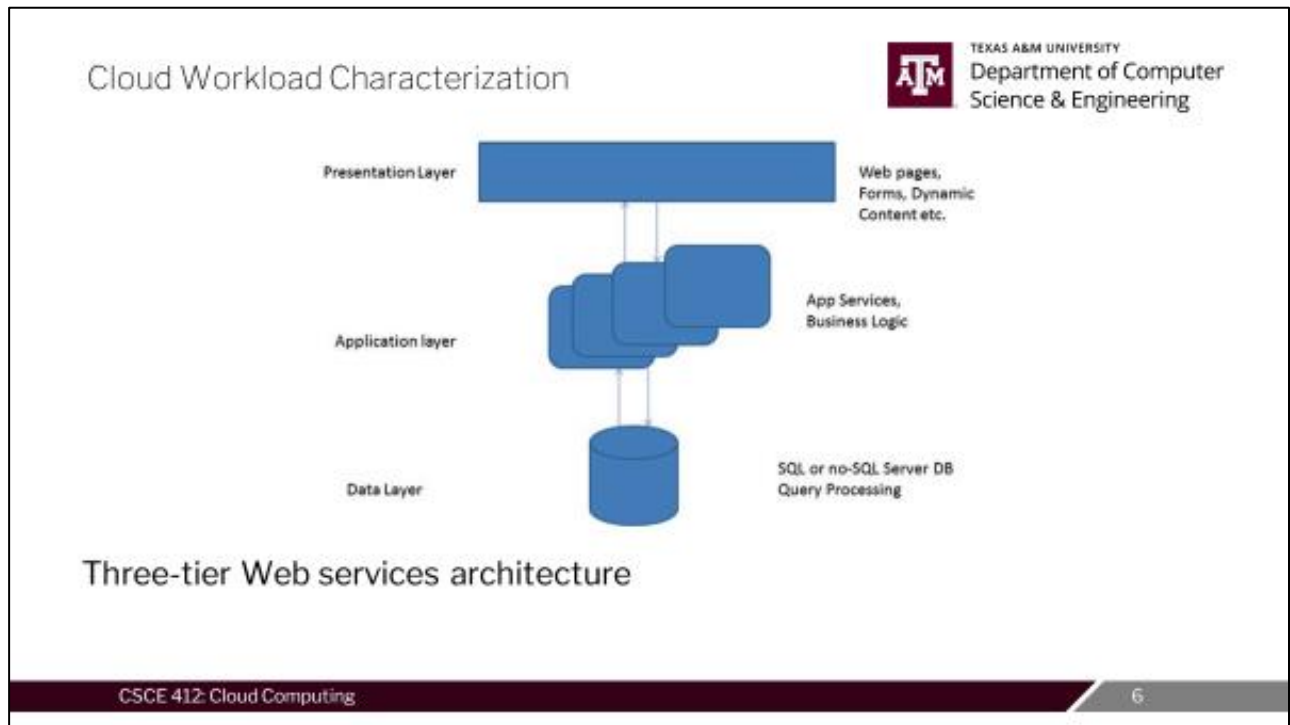
What do cloud providers do to optimize for client workloads, especially when the load is dynamic?

- Build a profile to record historical utilization and generate a pattern to predict near-future usage.
- Dispersion of users who frequently have high utilization across different servers.
- Dividing workloads into multiple VMs, running each VM on physical servers that are more equipped to handle the tasks, and devising an interconnect between them.

But what if utilization is more dynamic? How do providers address changing needs? This is a very big reason why people will hesitate to migrate to the cloud. Since it is hard to plan for the utilization of resources, it also becomes very difficult to allocate those resources while leaving adequate headroom. Despite the difficulty, cloud providers have devised a few ways to address this problem. One way to address this issue is to migrate the instance to another machine.

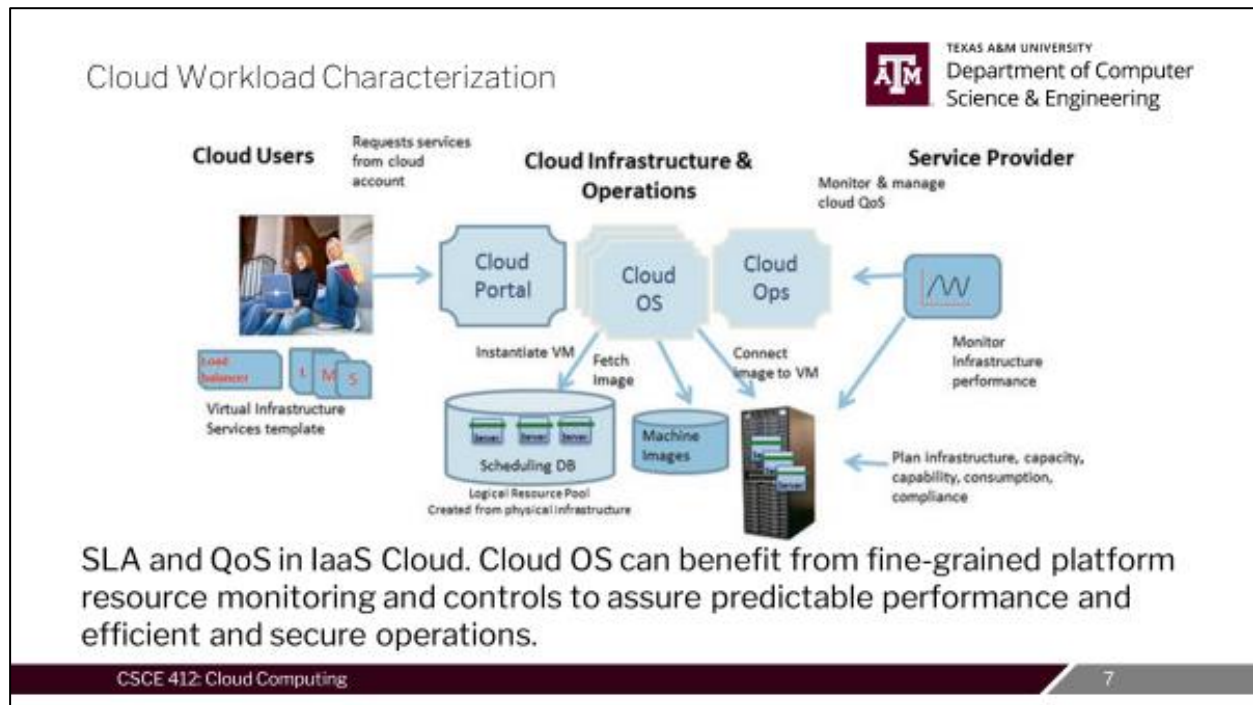
However, this tends to be very difficult to accomplish for instances that are hosting real-time processes that cannot incur any kind of interruptions. Another option would be to continually track an instance's utilization and predict its needs based on historical trends to preemptively allocate resources. However, when several instances allocate preemptively, a server may run out of physical resources.

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This leads to the option of dispersing users or instances that have frequently high utilization across different physical systems to ensure physical capacity isn't exceeded.

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However, if possible, dividing a workload into more specific, smaller workloads is another option. The divided workloads can then be made into multiple smaller workloads that can be put into their own smaller VMs, where each VM can then be hosted on physical servers that are more equipped to handle the tasks. Following this division, the provider then devises an interconnect between the instances to provide the service as normal to the users. Using these solutions allows providers to better serve dynamic workloads.

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In this module, we learned about some of the different cloud workloads, metrics to monitor for optimizing utilization, and the different ways providers address dynamic workload needs.



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